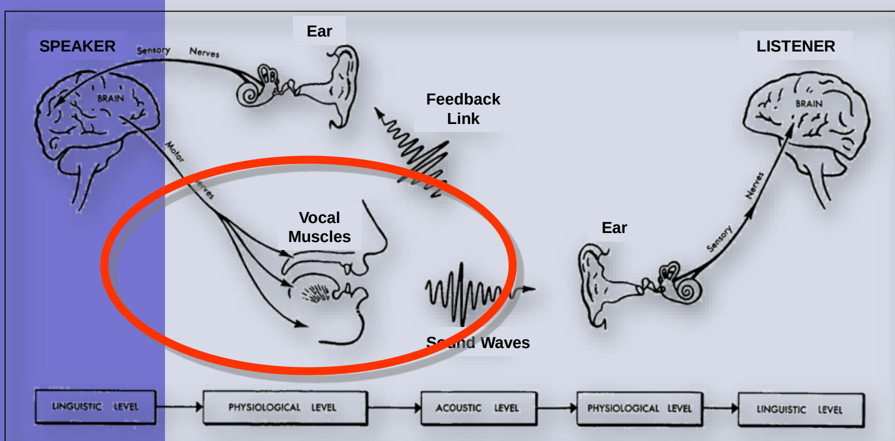
COM3502-4502-6502
SPEECH PROCESSING

Lecture 10
Speaking

1

The Speech Chain



Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*: New York: Anchor Press.

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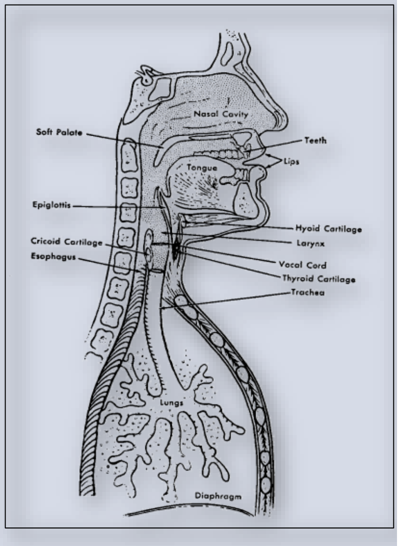
The Human Vocal Organs

The vocal apparatus has evolved for …
- breathing
- eating
- vocalising
- speaking



Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*: New York: Anchor Press.

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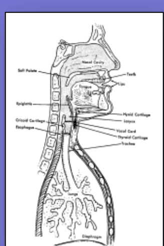
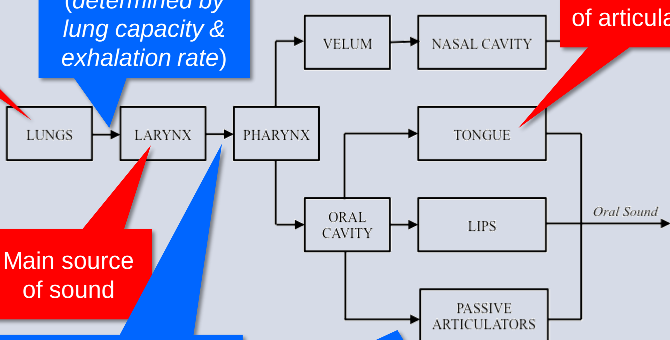
Mammalian Vocalisation

Main source of energy

Airflow (*determined by lung capacity & exhalation rate*)

Main source of articulation





Main source of sound

Harmonic-rich 'buzz' (*fundamental frequency determined by body size*)

Vocal tract resonant frequencies (*determined by body size & position of articulators*)

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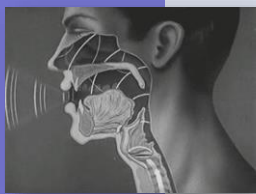
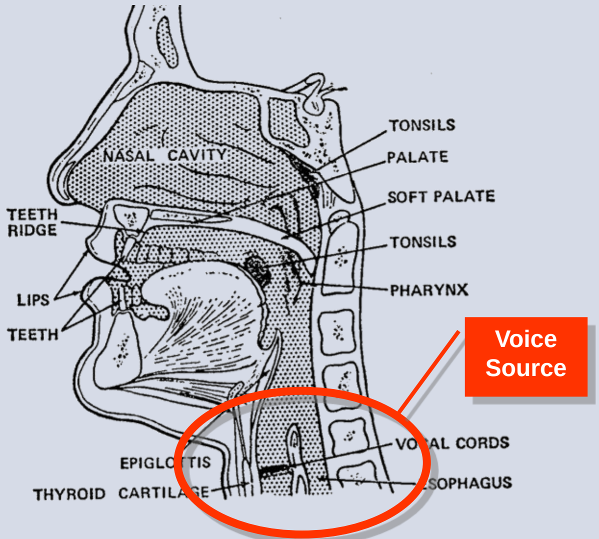
The Human Vocal Tract





Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*: New York: Anchor Press.

The Human Larynx

https://youtu.be/-XGds2GAvGQ

The Human Larynx

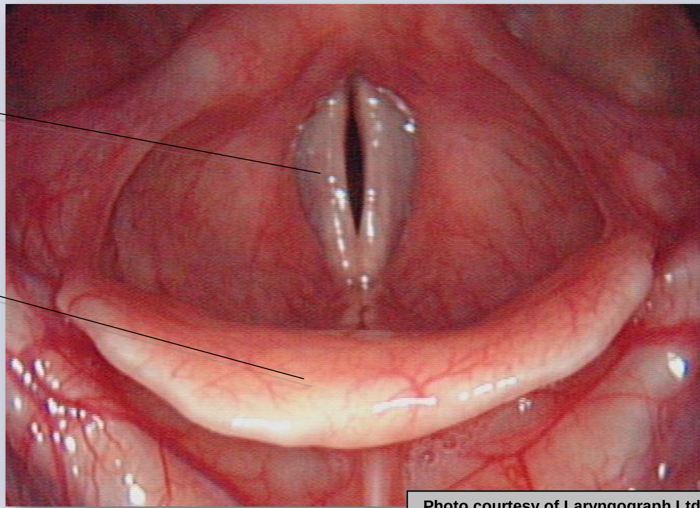
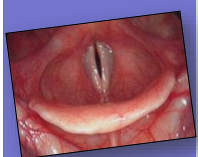


Photo courtesy of Laryngograph Ltd.

Voice ‘Source’



- Air pressure from the lungs builds up behind closed ‘vocal folds’ (often called ‘vocal cords’)
- The vocal folds are repeatedly forced apart and pulled together again, producing a series of small pulses of air
- This modulation of the airstream is known as ‘phonation’
- The tension in the muscles attached to the vocal folds determines their rate of vibration and hence the ‘fundamental frequency’ (F_X *or* F_0) of the speech waveform

- The fundamental frequency contributes to the perceived ‘pitch’ of the voice
- Because the vibration is not a pure sine wave, there is energy at frequencies that are multiples of the fundamental frequency (*known as the ‘pitch harmonics’*)

Two Cycles of Vocal Fold Vibration

Laryngograph

AVI

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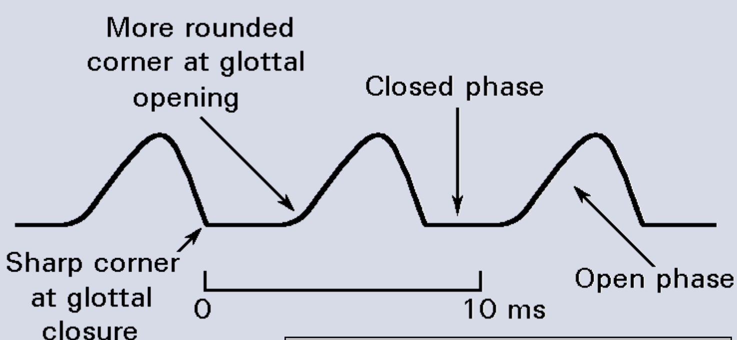
Air-Flow Through Glottis

The acoustic output from the vocal tract is initiated by the closure of the vocal folds



Taken from: Holmes, J. N., & Holmes, W. (2002). *Speech Synthesis and Recognition*: Taylor & Francis.

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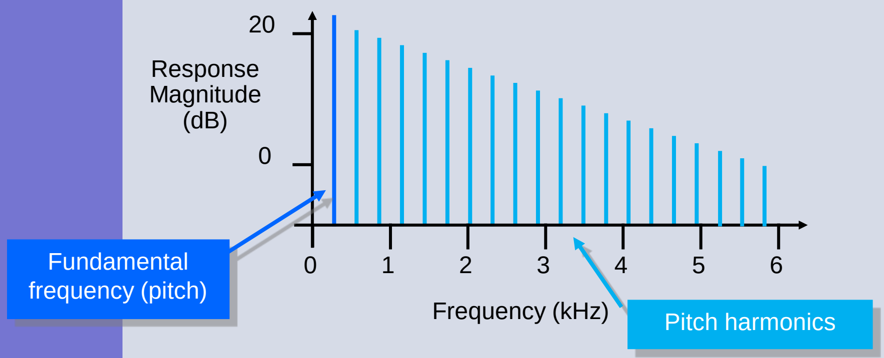
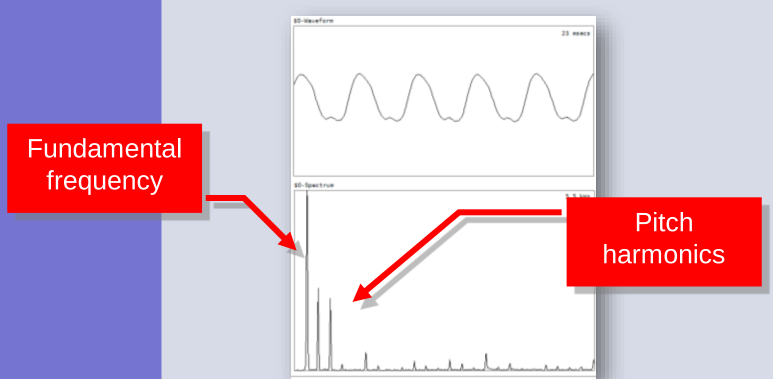
The Excitation Spectrum



Analysing the Voice Source



Microphone placed on the larynx

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Artificial Larynx / Electrolarynx
Charger Included
SolaTone™
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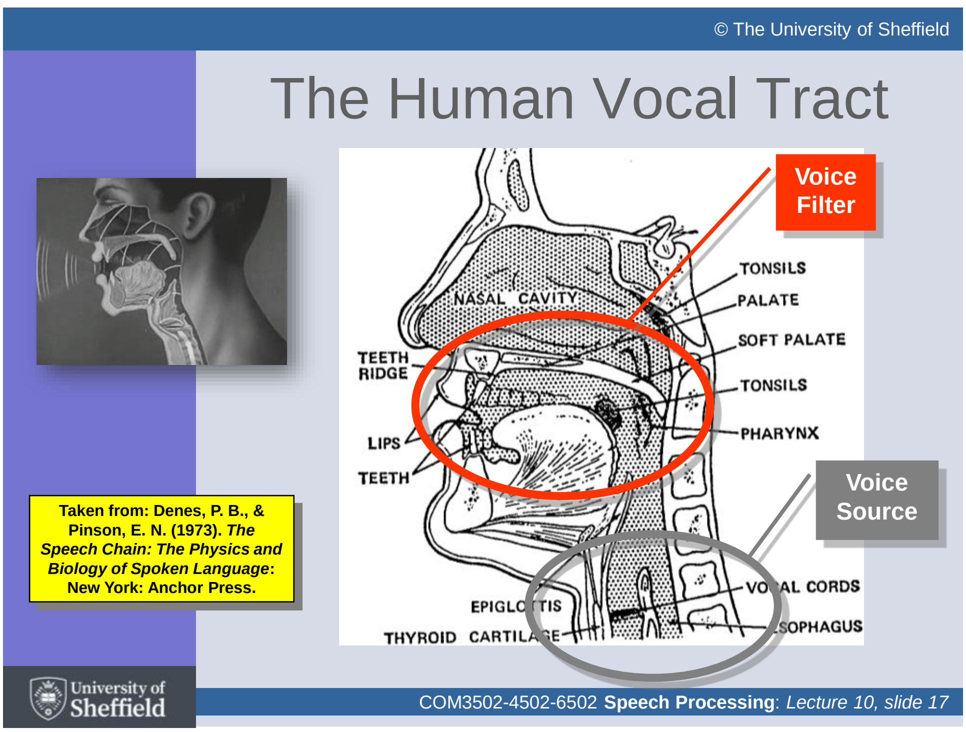

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The Human Vocal Tract
Voice Filter
NASAL CAVITY
TONSILS
PALATE
SOFT PALATE
TEETH RIDGE
TONSILS
PHARYNX
LIPS
TEETH
Voice Source
THYROID
Taken from: Denes, P. B., & Pinson, E. N. (1973). The Speech Chain: The Physics and Biology of Spoken Language: New York: Anchor Press.
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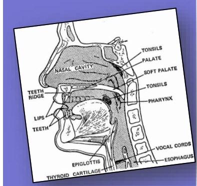
Voice 'Filter'



- The vocal tract forms a '**resonator**' with a complex shape

- Resonances are known as '**formants**'
- Speech is produced by using the '**articulators**' to change the shape of the vocal tract, hence modifying its resonant characteristics
- Different configurations of the vocal tract enhance some of the harmonics of the pitch, and suppress (*damp*) others
- The principal articulator is the **tongue**, but the jaw, lips, soft palate and teeth are also involved

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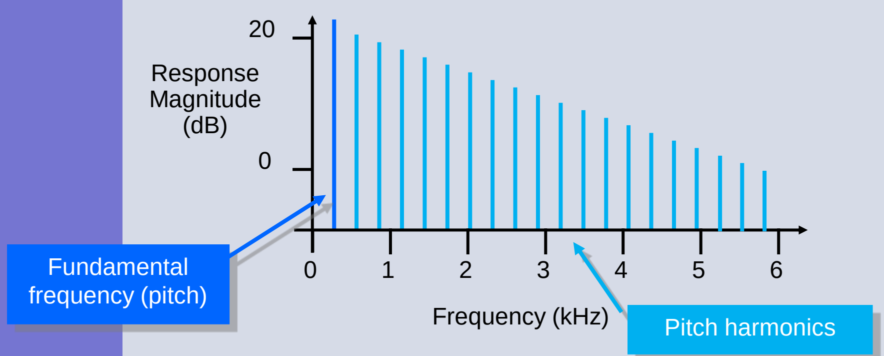
The Excitation Spectrum



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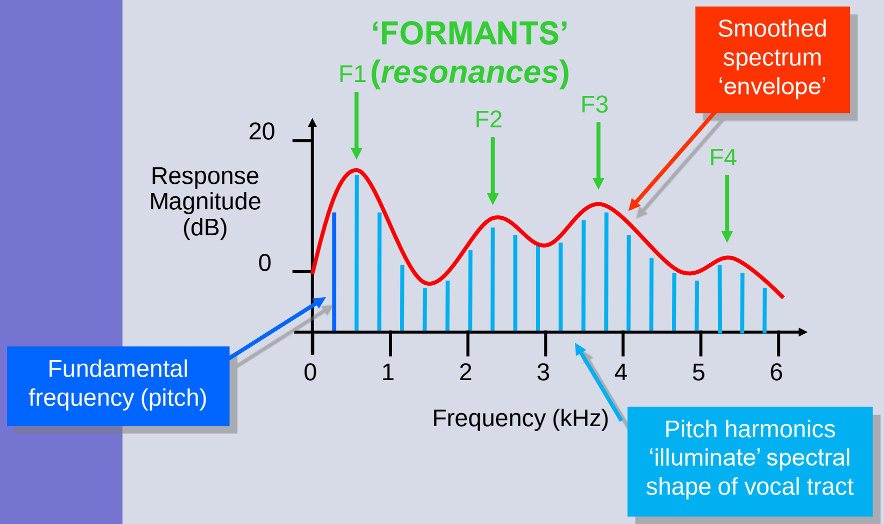
The Speech Spectrum



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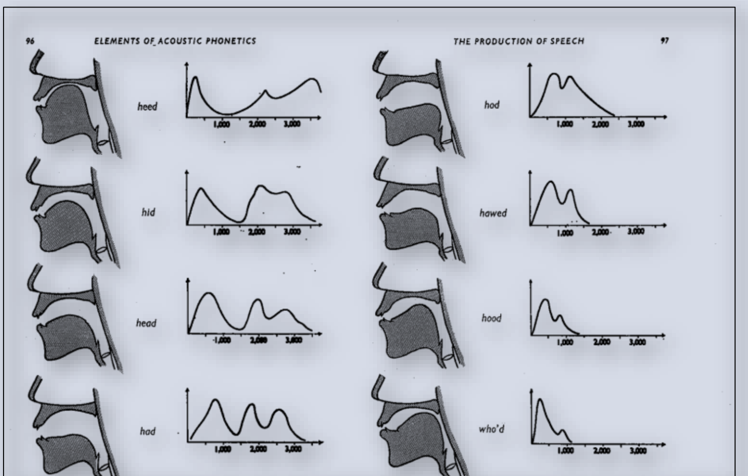
Vocal Tract Shape and Spectra



Taken from: Ladefoged, P. (1962). *Elements of Acoustic Phonetics*: London: University of Chicago Press.

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A Static Vocal Tract

Rhys Prosser

electrolarynx

+

3D-printed tubes

The Moving Vocal Tract

Magnetic Resonance Imaging (MRI)

University of Southern California
USC
Speech Analysis and Interpretation Laboratory
USC Viterbi
School of Engineering

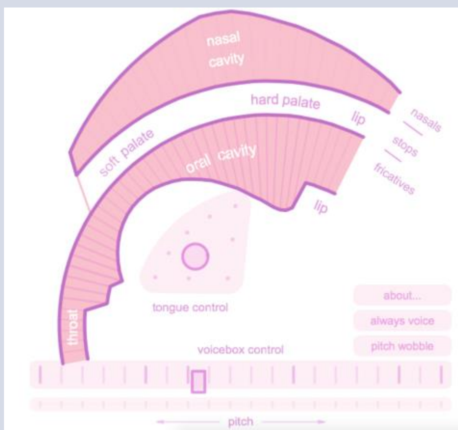
http://sail.usc.edu/span/index.php

The Moving Vocal Tract

UCL

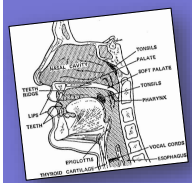
https://youtu.be/yGV8az8npZU

The Moving Vocal Tract



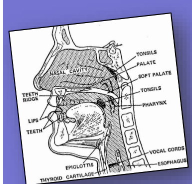
https://dood.al/pinktrombone/

Sound Source



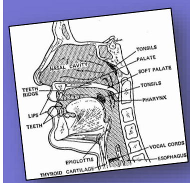
- Question …
  – is the larynx the only source of sound in the human vocal tract?
- Answer, no …
  – sound can be generated anywhere where there is a partial constriction *(e.g. "sh")*
  – or by exciting a resonance *(e.g. a whistle)*
  – or by vibrating an articulator *(e.g. the tongue)*
  – or by releasing a blockage *(e.g. the lips)*

Types of Speech Sound



- A 'voiced' sound is one in which the vocal cords are vibrating
- An 'unvoiced' sound is one in which the vocal cords are not vibrating
- A 'fricative' sound results from a turbulent air flow at a constriction
- A 'plosive' sound occurs after a blockage is released

Sound Exercise



Which of the following English words <u>end</u> in a ‘voiced’ sound?

- bus x
- breathe ✓
- has ✓
- off x
- buzz ✓
- breath x
- rule ✓
- of ✓

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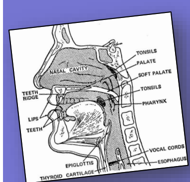
Sound Exercise



Which of the following English words <u>end</u> in a ‘fricative’ sound?

- bus ✓
- breathe ✓
- hat x
- off ✓
- bun x
- teeth ✓
- rule x
- of ✓

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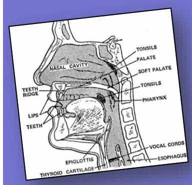
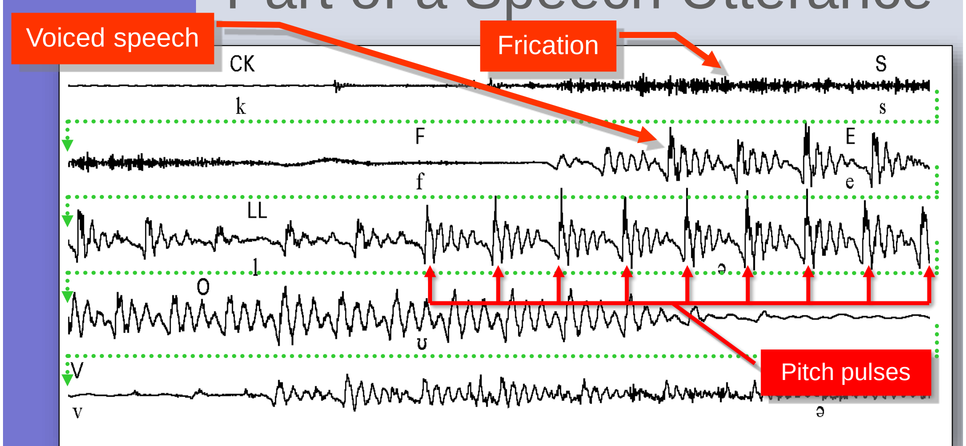
Sound Exercise



Which of the following English words start with a ‘**plosive**’ sound?

- bus ✓
- give ✓
- has x
- sat x
- pull ✓
- teeth ✓
- rule x
- cough ✓

Part of a Speech Utterance



“briCKS FELL OVER”

Taken from: Holmes, J. N., & Holmes, W. (2002). *Speech Synthesis and Recognition*: Taylor & Francis.

This slide set has covered …

- The human vocal tract
- The larynx
- Generating pitch pulses
- Voice 'source' and voice 'filter'
- The speech spectrum
- Resonances/formants
- Generating speech sounds
- Types of speech sounds

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Any Questions ?

Raise during lecture or post in the Blackboard Discussion group

Next time …

Hearing